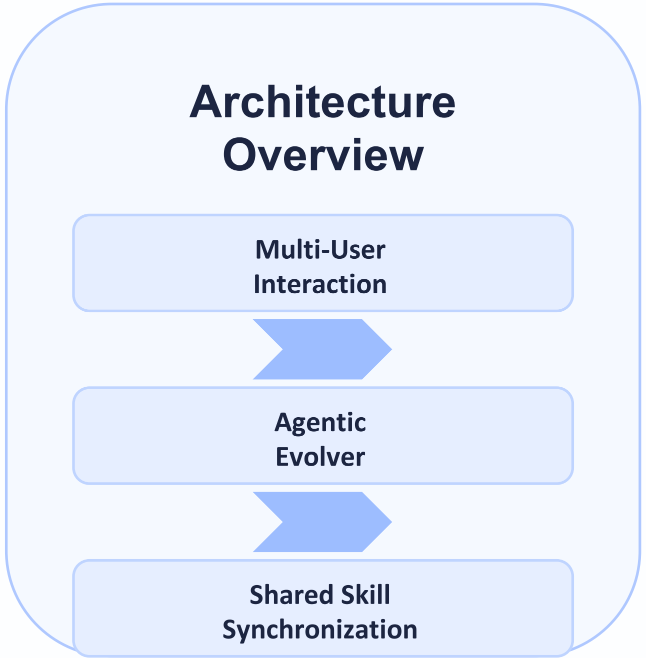
SkillClaw: Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Let Skills Evolve Collectively with Agentic Evolver

SkillClaw enables LLM agent skills to continuously evolve by aggregating cross-user interaction trajectories and applying agentic, evidence-driven skill updates.



Source: arXiv:2604.08377

Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently

Static Paradigm

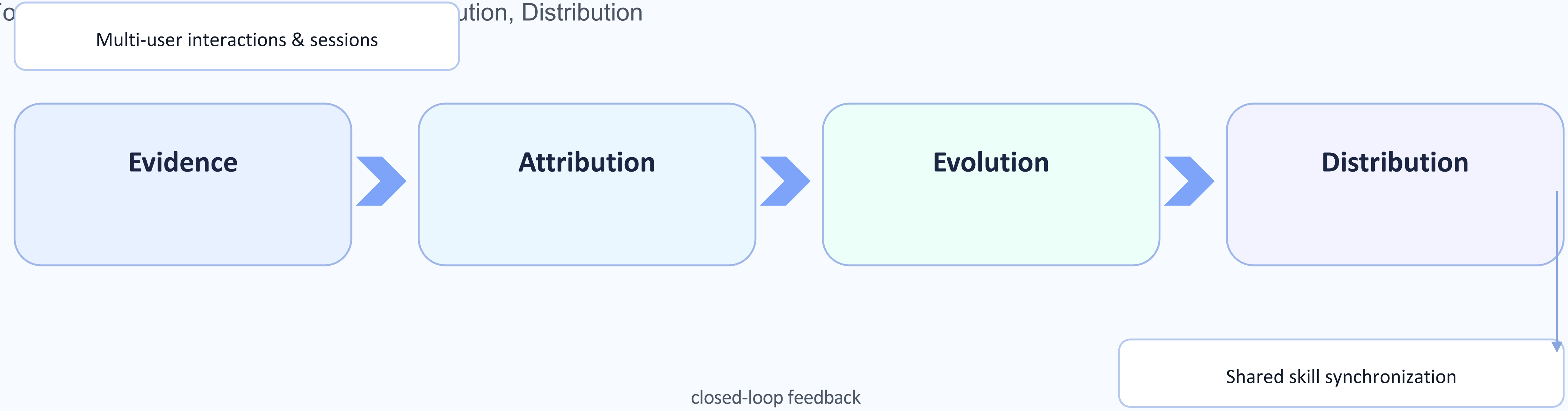
- Skills are manually installed and fixed
- Runtime failures are not converted into updates
- Different users independently rediscover workarounds
- No shared memory across deployments

Evolving Paradigm (SkillClaw)

- Interaction traces become evidence
- Agentic evolver diagnoses root causes
- Validated skill mutations are synchronized
- Shared ecosystem improves continuously

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution

Fo ution, Distribution



Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

1 Evidence Analysis

Aggregates multi-user trajectories and surfaces recurring success/failure patterns.

2 Attribution

Diagnoses root causes and isolates which existing skill behavior should change.

3 Evolution

Proposes candidate skill mutations via open-ended agent reasoning and tool use.

Key idea: evolution is agentic and open-ended, not a predefined update rule engine.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Compatible Claw-style ecosystems

OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw

Diverse contexts produce complementary signals (success envelopes + failure boundaries).

Why multi-user matters

Single-user traces rarely separate generalizable improvements from idiosyncratic fixes.

Aggregation increases evidence quality for reliable skill mutation decisions.

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

Date	Outcome	Rationale
Day 1	**Rejected**	Candidate overfit to narrow prompt style; failed broader prompts.
Day 2	**Rejected**	Improved speed but reduced output consistency across modalities.
Day 3	**Rejected**	Formatting mutation caused regressions in downstream composition.
Day 4	**Accepted**	Balanced generation + assembly constraints; passed validator checks.
Day 5	**Rejected**	Extra mutation did not surpass accepted baseline performance.
Day 6	**Rejected**	Context compression change removed useful creative control signals.

Conservative verification behavior: Accepted 1/6 candidates during 6-day evolution window.

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Date	Outcome	Rationale
Day 1	Accepted	Introduced robust auth fallback path for standard push failures.
Day 2	Rejected	Retry mutation duplicated auth prompts and increased friction.
Day 3	Accepted	Refined credential detection and reduced unnecessary retries.
Day 4	Rejected	Aggressive timeout shortened valid network recovery window.
Day 5	Accepted	Improved failure classification for token-vs-SSH mismatch cases.
Day 6	Rejected	Further edits plateaued; no validator gain over current best skill.

Iterative refinement when evidence supports updates: Accepted 3/6 candidates during 6-day evolution window.

Key Takeaways: Three Properties That Distinguish SkillClaw from Existing Systems

Implications for continuously improving agent ecosystems

1. **Collective evolution**

   Individual interactions contribute to a shared, continuously improving skill ecosystem.

2. **Fully automatic**

   Runtime interaction drives skill updates without manual curation or explicit user intervention.

3. **Agentic evolution paradigm**

   Open-ended reasoning over evidence enables context-aware skill mutations beyond fixed rules.

Implication: SkillClaw operationalizes continuously improving agent systems by turning distributed user experience into validated shared capabilities.